

Chapter 1

A (very) brief look at number systems

A precise and rigorous description and construction of the various number systems would require a whole course in itself. We shall not be examining these technical issues in detail in these notes but we may come across some of them from time to time. For the time being we shall introduce and agree on some basic terminology for the number systems we shall use, and indicate some of the technical detail that distinguishes the real numbers from the rational numbers.

- The *Natural numbers*, $\mathbb{N} = \{1, 2, 3, \dots\}$. These are the basic counting numbers¹.
- The *Integers*, $\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$. These consist of the natural numbers and their negatives, together with 0.
- The *Rational numbers*, $\mathbb{Q} = \left\{\frac{m}{n} : m, n \in \mathbb{Z}, m \neq 0\right\}$. These consist of all the fractions formed from integral numerators and denominators.
- The *Real numbers*, $\mathbb{R}$. A real number corresponds to a point on the number line (or continuum). They can be represented in many ways, for instance as infinite decimal expansions (or infinite digit expansions to any number base), as limits of infinite sequences of rational numbers, as infinite continued fractions,...
- The *Complex numbers*, $\mathbb{C} = \{a + bi : a, b \in \mathbb{R}\}$, where $i = \sqrt{-1}$.

If we disregard some technical details we can regard these number systems as being *nested*, i.e. the natural numbers are contained within the integers, which in turn are contained within the rational numbers, which in turn are contained within the real numbers. This is indicated symbolically as

$$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C} \subset \dots$$

Examples of real numbers which are not rational numbers (so called irrational numbers) are $\sqrt{2}$, $\sqrt{3}$, π , etc. The rationals and reals are pervasive (or

¹Some authors regard 0 as a natural number

dense) everywhere on the number line. In any interval on the number line, there are an infinite number of rational and irrational numbers.

Describing precisely what the difference is between the rational numbers and the real numbers is quite a technical matter. It comes down to a principle called the *least upper-bound principle*.

Least upper-bound principle

A system of numbers S has the least upper-bound principle if for every subset $A \subset S$ which is bounded above, there exists in S a *least* upper-bound for A .

That the rational numbers do not have this property can be seen from examples such as

$$A = \{x \in \mathbb{Q} : x^2 < 2\}.$$

The set A is clearly bounded above (by 3 for instance). But there is no least-upper bound for A in the rational numbers. There is a least upper-bound for A in the *real* numbers, namely $\sqrt{2}$. But this is not a rational number. The real numbers though do satisfy the least upper-bound principle.